

# Keyed-to-circular peg insertion

ManiSkill / PhysX · Panda manipulator, pd\_joint\_pos · motion-planning actions

Intervention  $c = [\Delta x, \Delta y, \Delta \psi]$  (task-local: lateral translation  $\times 2$ , axial yaw)

Frozen manifest: 39 conditions = 30 mixed (train) + 8 isolated (held-out) + 1 zero baseline

strict pass

**Single-seed  
phase switch**  
1 seed · 39 conditions

**39**

usable episodes

strict pass

**Five-seed  
replication**  
5 seeds  $\times$  39 conditions

**195**

usable episodes

no drops

**Few-shot sweep**  
 $N \in \{3, 5, 8, 10, 15, 20, 30\}$  · 5 seeds

**1,815**

model fits

complete

**TP-GMM matched  
few shot**  
 $N \in \{5, 10, 20, 30\}$  · matched cells

**80**

model fits

frozen

**Rotated axis  
( $Q_1$  /  $Q_2$ )**  
2 orientations  $\times$  2 seeds  $\times$  39

**232**

usable episodes

collected

**Circular symmetry  
(honest / placebo)**  
2 variants  $\times$  3 seeds  $\times$  39

**234**

usable episodes

**6 experiments · 700 usable physics rollouts · 1,895 offline model fits**

All rollouts executed in simulation (ManiSkill/PhysX). Failed attempts retained with condition id, attempt id, seed, stop reason, and source hashes.